

Diabetes Prediction

Observations and Comparison

After training and evaluating the models on the scaled test data, we can compare their performance based on accuracy and other metrics.

- **SVM with RBF kernel** achieved the highest accuracy among all models. This is expected because RBF kernel can handle non-linear relationships well, and the dataset likely has some non-linear patterns.
- **SVM with linear kernel** performed slightly lower but still reasonably well. It assumes a linear decision boundary.
- **SVM with polynomial kernel** (degree 3) gave similar accuracy to linear, but might be prone to overfitting if the degree is too high.
- **SVM with sigmoid kernel** performed the worst, which is common because sigmoid kernel is not always suitable for classification tasks and can behave like a two-layer neural network but often underperforms.
- **KNN (k=5)** performed comparably to the linear SVM, but

slightly lower than RBF. KNN's performance depends on the choice of k and the distance metric; with scaled features it works reasonably.

Overall, **SVM with RBF kernel** performed best on this dataset. However, note that we used default parameters. Hyperparameter tuning (e.g., C , γ for RBF, or k for KNN) could potentially improve results further.

Conclusion

Based on this comparison, the RBF SVM is the recommended model for this diabetes prediction task. The linear SVM and KNN are also viable alternatives but with slightly lower accuracy.